

Ridge Regression with Gradient Descent

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Where this fits: Pipeline map step 8 (Model). See the [Course map](#).



- **Builds on:** Standardization ([Note ML-009](#)); Multiple linear regression ([Note ML-052](#)); Normal equation ([Note ML-053](#)); Gradient descent ([Note ML-056](#)); Bias-variance trade-off ([Note ML-061](#)); Regularisation ([Note ML-062](#)).
- **Leads to:** Elastic Net ([Note ML-068](#)).
- **Compare with:** Lasso regression ([Note ML-066](#)); L1 and L2 regularisation in neural networks ([Note DL-026](#)).

1. Overview

Key point: Ridge can also be trained with gradient descent. The gradient is the linear regression gradient plus λw , so each step also pulls the coefficients a little towards 0.

The previous Note found the Ridge coefficients in one step with a formula. Like plain linear regression, **Ridge regression** (G-1691) can instead be trained with **gradient descent** (G-862): start somewhere, and repeatedly step downhill on the loss.

Here a **feature** (G-772) is an input variable (one column of the data table), an **observation** (G-1374) is one record (one row), and the **target** (G-1949) y is the value we predict. Gradient descent is useful when there are many features, because the formula needs the inverse of a large matrix (the gradient descent Notes). This Note derives the Ridge gradient, codes it from scratch, and shows the scikit-learn options.

2. The gradient

Key point: With a factor of $\frac{1}{2}$ in the loss, the gradient is $X^T X w - X^T y + \lambda w$.

2.1 The loss with a factor of $\frac{1}{2}$

Key point: Multiplying the loss by $\frac{1}{2}$ does not move its minimum; it only removes a 2 from the derivative.

The Ridge loss from the previous Note, multiplied by $\frac{1}{2}$:

$$L = \frac{1}{2}(Xw - y)^T(Xw - y) + \frac{1}{2}\lambda w^T w$$

Halving every value of the loss leaves the lowest point at the same w . The $\frac{1}{2}$ just makes the derivative tidier, as the 2 from differentiating a square cancels the $\frac{1}{2}$.

2.2 Expanding and differentiating

Key point: The same expansion as the previous Note, then the same derivative rules.

Expanding the product as in the normal equation Note:

$$L = \frac{1}{2} \left(w^\top X^\top X w - 2w^\top X^\top y + y^\top y \right) + \frac{1}{2} \lambda w^\top w$$

Differentiating with respect to w :

$$\frac{\partial L}{\partial w} = X^\top X w - X^\top y + \lambda w$$

The first two terms are the linear regression gradient. The Ridge penalty adds only λw .

Extra: Setting this gradient to zero gives $(X^\top X + \lambda I)w = X^\top y$, the formula of the previous Note. Gradient descent walks towards the same answer step by step instead of jumping to it.

2.3 The update rule

Key point: New w = old w minus the learning rate times the gradient.

As in every gradient descent Note, all coefficients are updated together:

$$w_{\text{new}} = w_{\text{old}} - \eta \left(X^\top X w_{\text{old}} - X^\top y + \lambda w_{\text{old}} \right)$$

where η is the **learning rate** (G-1068). The first entry of w is the intercept, which is not penalised. So, as before, the λw term uses $\lambda I w$ with the top-left entry of I set to 0.

Extra: Rearranging the update gives $w_{\text{new}} = (1 - \eta\lambda) w_{\text{old}} - \eta (X^\top X w_{\text{old}} - X^\top y)$. So every step first shrinks the coefficients by the factor $1 - \eta\lambda$, then takes the ordinary linear regression step. The per-step shrinking is why the L2 penalty is also called **weight decay** (G-2108), the name used for neural networks (ESL §3.4.1).

Figure 1 splits the first step of the one-feature example of Section 3 ($\lambda = 100$, $\eta = 0.005$) into its two parts. Watch the red arrow: the shrink halves the slope before the ordinary step is taken.

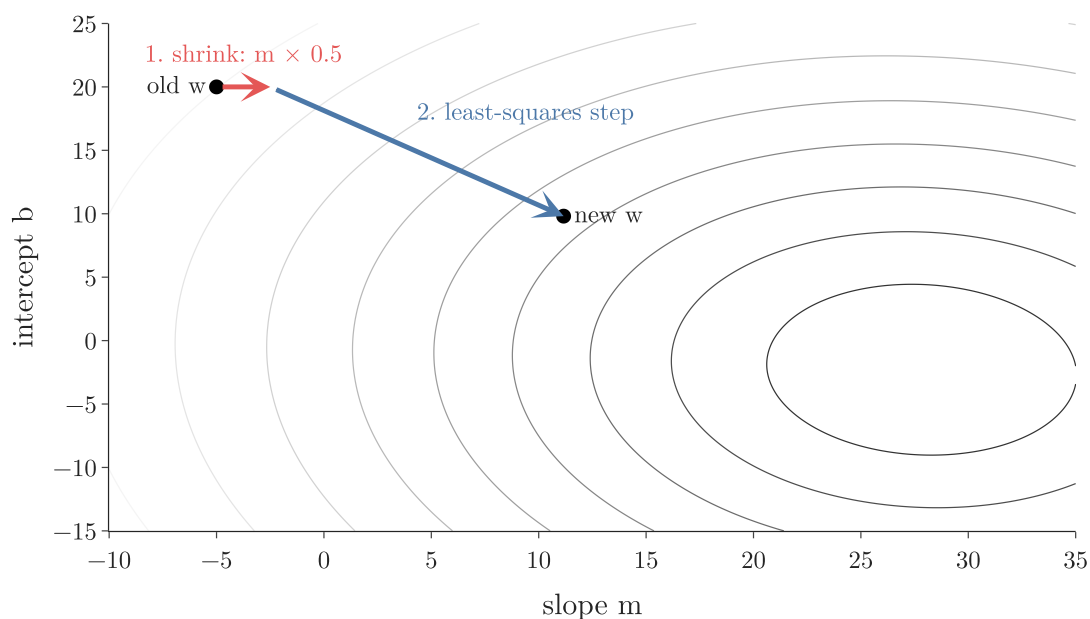


Figure 1: One Ridge step in two parts, on the squared-error contours: shrink the slope by $1 - \eta\lambda = 0.5$ (red), then take the ordinary least-squares step (blue). The intercept is not shrunk

3. Seeing the paths

Key point: Gradient descent finds the bottom of whichever loss it is given. With $\lambda = 100$, the bottom sits at a smaller slope.

Figure 2 runs gradient descent on the one-feature example of the previous Notes for three values of λ , from the same start point ($m = -5$, $b = 20$, learning rate 0.005). The contours belong to the plain squared error, so only $\lambda = 0$ heads for their centre. Watch where each path stops: the larger λ , the closer to $m = 0$, and the flatter its line on the data (right).

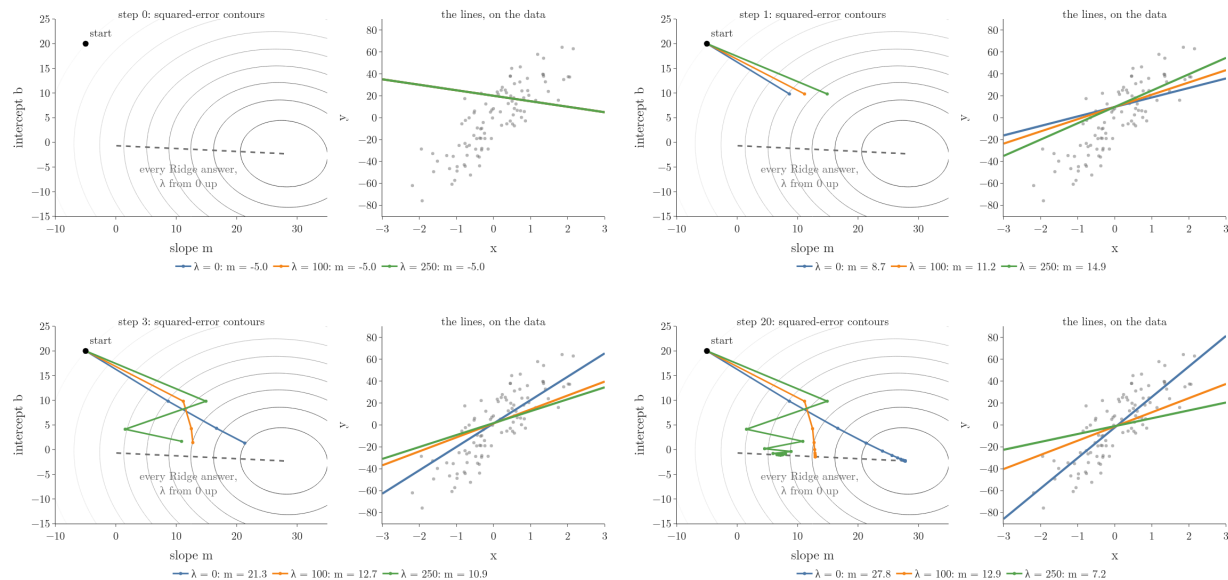


Figure 2: Gradient descent with $\lambda = 0, 100$ and 250 from the same start. Left: the paths on the squared-error contours; the dashed curve holds every Ridge answer as λ grows from 0. Right: each path's current line on the data

With $\lambda = 250$ the path zig-zags across the valley: the penalty makes the bowl so steep in the m direction that a step of 0.005 overshoots, and each overshoot is smaller than the last (the Extra in Section 4 gives the limit). Figure 3 draws the bowls themselves for $\lambda = 0$ and $\lambda = 100$ over 60 steps.

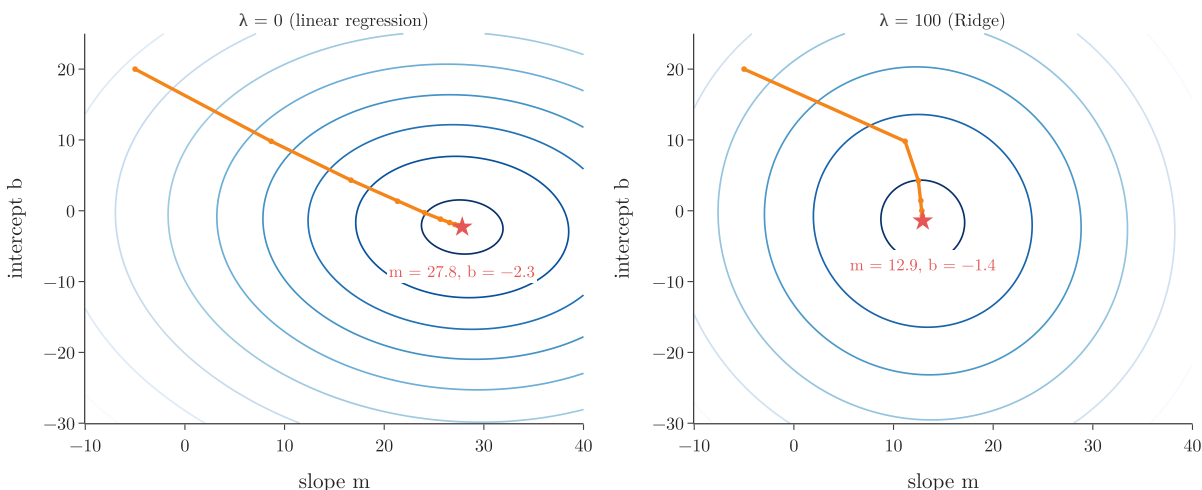


Figure 3: Gradient descent on the loss with $\lambda = 0$ and $\lambda = 100$

- $\lambda = 0$: the path ends at $m = 27.8, b = -2.3$, the linear regression answer.

- $\lambda = 100$: the penalty moves the bottom of the bowl to $m = 12.9$, $b = -1.4$. These values are exactly the Ridge values from the formula in the previous Note.

The penalty makes the bowl steeper in the m direction only: its curvature there, $\partial^2 L / \partial m^2 = \sum x_i^2 + \lambda$, grows from 87 to 187, while the b direction is unchanged.

4. Code from scratch

Key point: The loop computes the gradient with matrix products and updates all coefficients at once.

Python: Ridge with batch gradient descent.

```
class MyRidgeGD:
    def __init__(self, epochs, learning_rate, alpha):
        self.epochs, self.lr, self.alpha = (epochs,
                                             learning_rate, alpha)

    def fit(self, X, y):
        X = np.insert(X, 0, 1, axis=1)           # column of 1s
        w = np.insert(np.ones(X.shape[1] - 1), 0, 0.0)
        I = np.identity(X.shape[1])
        I[0, 0] = 0                               # no penalty on intercept
        for _ in range(self.epochs):
            grad = X.T @ X @ w - X.T @ y + self.alpha * I @ w
            w = w - self.lr * grad
        self.intercept_, self.coef_ = w[0], w[1:]
        return self

    def predict(self, X):
        return X @ self.coef_ + self.intercept_
```

On the diabetes data (test size 0.2, random state 4) with alpha 0.001, learning rate 0.005 and 500 epochs, test R^2 is 0.474 and the intercept 150.87.

Extra: This gradient adds up the errors of all observations instead of averaging them, so its size grows with the number of observations. The summed gradient is why the learning rate must be tiny here. The maths: write $H = X^T X + \lambda I$ and w^* for the exact answer. One update turns the error $w - w^*$ into $(I - \eta H)(w - w^*)$. Along each eigenvector of H with eigenvalue h , the error is multiplied by $1 - \eta h$ every step, so it shrinks only if $\eta < 2/h$ for the largest h . On this training set (353 observations) the largest eigenvalue is 353, so the limit is $\eta < 2/353 = 0.0057$. 0.005 is just below it; 0.006 is just above it, and the steps overshoot and grow: after 100 epochs the largest coefficient is about 15,000.

Figure 4 runs the code above at both learning rates. For about 50 epochs the two runs look alike; then the run with 0.006 bends upwards, because the overshoot along the steepest direction is multiplied by $|1 - 0.006 \times 353| = 1.12$ every step and finally dominates.

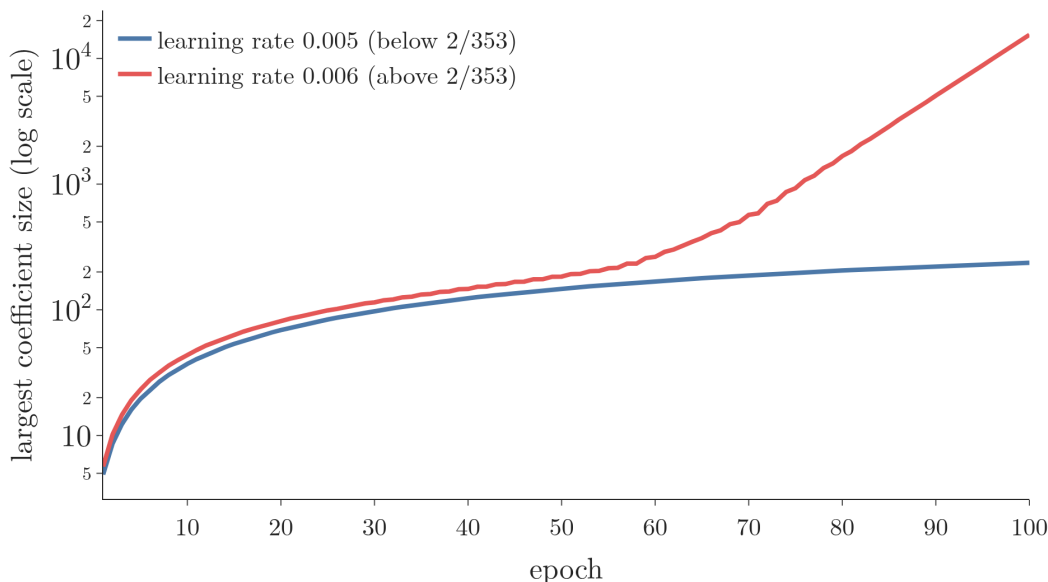


Figure 4: The largest coefficient of the from-scratch code per epoch on the diabetes data, alpha 0.001. Learning rate 0.005 (blue, below the limit $2/353$) settles; 0.006 (red, above it) grows to about 15,000 after 100 epochs. Log scale.

5. Gradient descent stops early

Key point: Stopping gradient descent early keeps the coefficients small, so **early stopping** (G-656) is a regulariser of its own, much like the ridge penalty.

Think of pouring water into an ice-cube tray: stop pouring early and every cube is only partly full. Gradient descent starts with small coefficients and fills them up step by step; stopping early leaves the slow ones small.

5.1 Early stopping rescues an overfitting model

Key point: On data where linear regression overfits badly, stopping early helps about as much as the ridge penalty.

We give the model 65 features (the 10 diabetes measurements plus their squares and pairwise products) and train on 200 patients, as in the [batch gradient descent Note](#) (section five). With so many features for so few observations, plain least squares has high variance, the case where shrinkage helps (ISL §6.2.1). Averaged over 50 random splits, test R^2 is:

Model	Test R^2
Linear regression (no regularisation)	0.06
Ridge, alpha chosen by cross-validation on the training data	0.44
Gradient descent without a penalty, stopped early	0.40

Both regularisers rescue the badly overfitting linear regression, and by a similar amount. For a linear model, stopping early and adding the ridge penalty are close cousins (Goodfellow et al. §7.8).

Extra: In the 65-feature experiment both models use standardised features. The stopping epoch is chosen on a validation part of the training data, and ridge’s alpha by cross-validation on the training data, so the test data is used only once. Early stopping is one of the most common regularisers in deep learning (Goodfellow et al. §7.8).

5.2 Why the slow directions stay small

Key point: Gradient descent creeps along flat directions of the loss bowl, so after a few hundred steps the coefficients in those directions are still small.

Back on the 10 diabetes features (353 training observations), the exact Ridge answer from the previous Note scores test R^2 0.463. Figure 5 tracks gradient descent for up to a million epochs.

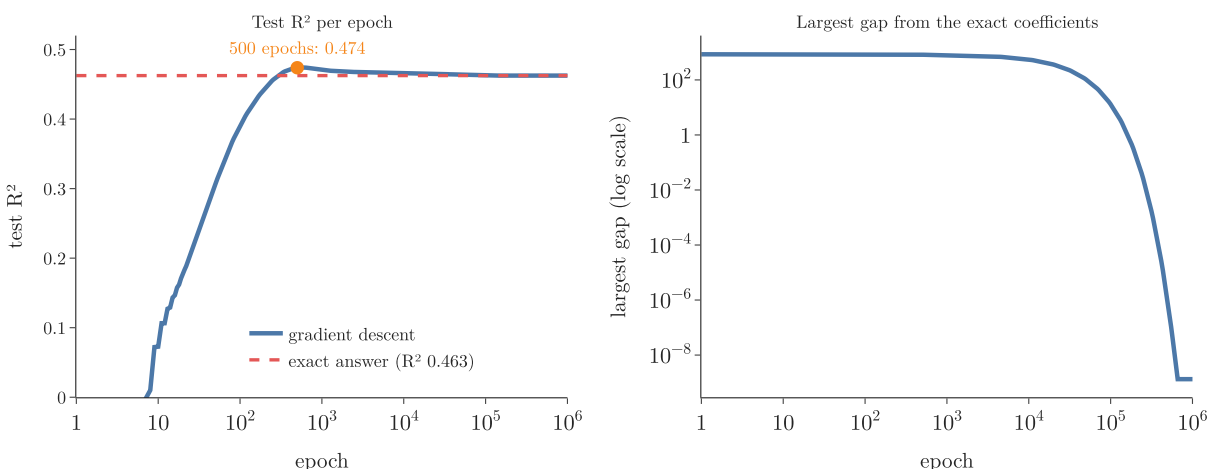


Figure 5: Test R^2 and distance from the exact answer, per epoch

Epochs	Test R^2	Largest gap from the exact coefficients
100	0.390	897
500	0.474	823
50,000	0.464	104
1,000,000	0.463	0.0

- **Left:** test R^2 rises, peaks around 500 epochs, then settles at the exact answer’s 0.463.
- **Right:** the coefficients need about a million epochs to reach the exact values.

The slow direction comes from s_1 and s_2 , which are strongly correlated (the [first Ridge Note](#)). The bowl is almost flat along “raise s_2 , lower s_1 ”, so gradient descent creeps along it, and

stopping early leaves the coefficients in that direction smaller (Goodfellow et al. §7.8). With 353 observations least squares barely overfits (the first Ridge Note), so on this data the size of the test- R^2 change says little; the 65-feature experiment above is the real test.

Extra: The numbers behind this. Along an eigenvector of $H = X^T X + \lambda I$ with eigenvalue h , the error shrinks by $1 - \eta h$ per step, so it needs about $1/(\eta h)$ steps. The smallest eigenvalue here is 0.0083, giving about $1/(0.005 \times 0.0083) \approx 24,000$ steps, and its eigenvector is mostly s_1 (-0.71) and s_2 ($+0.55$). After 500 epochs, almost the whole gap from the exact answer (1,146 of it) lies along this one direction, and the coefficient vector is about half as long as the exact one (774 against 1,455).

6. Ridge with gradient descent in scikit-learn

Key point: Use `SGDRegressor` with `penalty="l2"`, or `Ridge` with an iterative solver.

Python: Two scikit-learn options.

```
from sklearn.linear_model import SGDRegressor, Ridge

sgd = SGDRegressor(penalty="l2", alpha=0.001, max_iter=500,
                   eta0=0.1, learning_rate="constant",
                   random_state=0)
sgd.fit(X_train, y_train)
sgd.score(X_test, y_test)           #  $R^2$  0.450

ridge = Ridge(alpha=0.001, max_iter=500, solver="sparse_cg")
ridge.fit(X_train, y_train)
ridge.score(X_test, y_test)        #  $R^2$  0.463
```

- **SGDRegressor** (G-1783; the stochastic gradient descent Note) adds the Ridge penalty with `penalty="l2"`. Its `alpha` is the penalty strength.
- **Ridge** has a `solver` (G-1836) setting. `"cholesky"` and `"svd"` use the formula directly. `"sparse_cg"`, `"lsqr"`, `"sag"` and `"saga"` reach the answer step by step, and `max_iter` limits the steps.

`Ridge` has no learning-rate setting at all, even with an iterative solver (scikit-learn docs, `Ridge`). Here `sparse_cg` reaches the exact answer, 0.463.

Extra: `SGDRegressor` averages the loss over the observations, while `Ridge` adds it up. The `SGDRegressor` objective is (scikit-learn user guide §1.5.8) $\frac{1}{n} \sum \frac{1}{2} (y_i - \hat{y}_i)^2 + \alpha \cdot \frac{1}{2} \|w\|^2$. Multiplying by $2n$ gives $\sum (y_i - \hat{y}_i)^2 + n\alpha \|w\|^2$, the `Ridge` loss with `alpha` $n\alpha$. So the same `alpha` means a stronger penalty in `SGDRegressor`: its `alpha` equals `Ridge`'s `alpha` divided by the number of observations. With 353 training observations, `SGDRegressor(alpha=0.001)` matches `Ridge(alpha=0.353)`: run long, its coefficients land within 3 of that model's, but up to 840 away from `Ridge(alpha=0.001)`.

Figure 6 shows the three sets of coefficients side by side. The orange bars of

SGDRegressor(alpha=0.001) sit on the blue bars of Ridge(alpha=0.353), not on the grey bars of Ridge(alpha=0.001); the biggest gap is in s1.

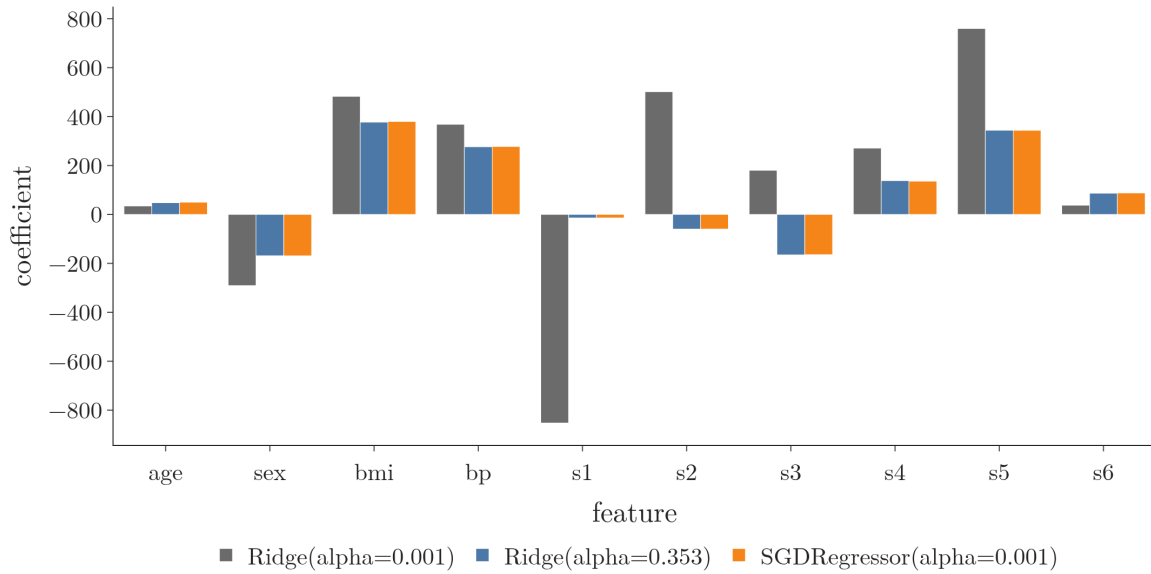


Figure 6: Coefficients on the diabetes training data (353 observations): Ridge with alpha 0.001 (grey) and 0.353 (blue), and SGDRegressor with alpha 0.001 run for 50,000 epochs (orange). The orange bars match the blue ones.

7. Summary

Method	How it finds w	Test R^2 here
Formula (<code>Ridge</code> , cholesky)	one step, $(X^T X + \lambda I)^{-1} X^T y$	0.463
<code>Ridge</code> , <code>sparse_cg</code>	iterative solver	0.463
<code>MyRidgeGD</code> , 500 epochs	batch gradient descent, stopped early	0.474
<code>SGDRegressor</code> , l2	stochastic gradient descent	0.450

- The Ridge gradient is $X^T X w - X^T y + \lambda w$: the linear regression gradient plus λw .
- Each update shrinks the coefficients by $1 - \eta \lambda$ first, hence the name weight decay.
- The intercept is left out of the penalty.
- Stopping gradient descent early also regularises: with many features for few observations it rescues an overfitting fit about as well as ridge.
- In scikit-learn: `SGDRegressor(penalty="l2")` or `Ridge` with an iterative solver.

8. Sources

Built from

- CampusX, “Ridge Regression Part 3 | Gradient Descent | Regularized Linear Models”, YouTube, https://www.youtube.com/watch?v=FcI_wwMp8G8

Other references

- **ESL**: Hastie, T., Tibshirani, R. and Friedman, J. *The Elements of Statistical Learning*, 2nd ed. Springer, 2009. Section 3.4.1, p. 63.
- **Goodfellow et al.**: Goodfellow, I., Bengio, Y. and Courville, A. *Deep Learning*. MIT Press, 2016. Section 7.8, “Early Stopping”.
- **scikit-learn docs**: `sklearn.linear_model.Ridge`; user guide Section 1.5.8, “Mathematical formulation” (SGD), scikit-learn 1.9.
- **ISL**: James, G., Witten, D., Hastie, T. and Tibshirani, R. *An Introduction to Statistical Learning*, 2nd ed. Springer, 2021. Section 6.2.1, p. 241.

9. Key terms

Term	Meaning
Weight decay	Another name for the L2 penalty: each gradient step shrinks the coefficients by a fixed factor
Solver	The method a scikit-learn model uses to find its coefficients
penalty	<code>SGDRegressor</code> setting that adds a regularisation penalty, such as “l2” for Ridge